

AIRLINE FLIGHT OPERATIONS & DELAY ANALYSIS

Final Project Report

Business Intelligence & Data Analytics Case Study

Project	Airline Flight Operations & Delay Analysis
Dataset	Maven Analytics Data Playground — Airline Flight Delays (U.S. DOT, 2015)
Period	Calendar Year 2015
Records	5,819,079 flight records
BI Platform	Microsoft Power BI
Techniques	Power Query / M, DAX, dimensional modeling, KPI validation
Deliverable	Interactive Executive Overview dashboard + supporting project documentation
Version	Final — August 2026

This report documents the project as it was actually implemented. It intentionally distinguishes the original analytical requirements from the final delivered dashboard so that the portfolio artifact remains technically accurate and interview-defensible.

1. Executive Summary

This project analyzes U.S. commercial flight operations recorded during 2015 using the Maven Analytics Airline Flight Delays dataset, sourced from U.S. Department of Transportation reporting. The project transforms flight-level records into a structured Power BI model and an interactive Executive Overview dashboard.

The implemented dashboard covers 5,819,079 flights. The validated dashboard results used in this project are 2,086,896 delayed flights, 3,627,112 on-time flights, 89,884 cancelled flights, and a 35.9% overall delay rate under the project's implemented classification. Average flight duration is 137.01 minutes.

The analysis identifies monthly variation in delay performance, differences among the highest-delay-rate airlines, concentration of delayed-flight volume at major origin airports, delay severity patterns, the leading recorded delay causes, and the distribution of cancellation reasons.

The strongest strategic distinction is between delays and cancellations: accumulated delay minutes are led by Late Aircraft, Air Carrier, and National Air System categories, while Weather is the dominant recorded cancellation reason. This supports different operational responses for delay reduction and cancellation resilience.

1.1 Executive Takeaways

- **Operational scale:** The dataset contains more than 5.8 million flight operations.
- **Delay exposure:** 2.09 million flights are classified as delayed in the implemented dashboard logic, producing a 35.9% project delay rate.
- **Delay concentration:** Late Aircraft, Air Carrier, and National Air System are the leading recorded contributors to total delay minutes.
- **Cancellation concentration:** Weather represents 54.35% of cancellations in the dashboard.
- **Management focus:** Delay reduction should prioritize operational efficiency and delay propagation; cancellation resilience should emphasize weather preparedness and contingency planning.
- **Analytical caution:** Raw delayed-flight counts should not be used alone to label an airport or airline as 'worst' because operating volume and mix affect those counts.

2. Business Problem & Objectives

The business problem is the absence of a consolidated analytical view that allows stakeholders to understand flight punctuality, delay concentration, and cancellation patterns quickly. The project was designed to convert historical operational records into decision-ready BI outputs that can be explored by managers and analysts.

2.1 Objectives

Objective	Implemented project contribution
Measure operational performance	KPI cards for total flights, delayed flights, on-time flights, delay rate, cancellations, and average duration.
Analyze delay patterns	Monthly delay-rate, delay-volume, severity, and delay-cause analysis.
Compare airlines	Top 5 airlines by delay rate.
Identify airport concentration	Top 5 origin airports by delayed-flight volume.
Analyze cancellations	Cancellation volume and reason mix.
Provide self-service BI	Interactive month and airline slicers with cross-filtering behavior.

2.2 Project Scope — Final Implemented State

The original project documentation defined a broader analytical target covering executive, airline, airport/route, and delay-cause views. The final implemented Power BI deliverable was intentionally narrowed to a single, focused Executive Overview dashboard rather than expanding the page count simply to satisfy the original design blueprint.

The final report therefore documents the **implemented dashboard** as the authoritative delivered BI artifact. The broader analytical areas remain valid as future extensions, but they are not presented as completed dashboard pages.

3. Domain Context

Each record represents one scheduled flight operation. The documented flight lifecycle progresses through scheduling, departure, en-route operation, arrival, and final disposition. A flight may be completed, cancelled, or diverted.

3.1 Key Terminology

Term	Project definition
Delay	Difference between scheduled and actual departure or arrival timing; the project uses a 15-minute threshold for delay classification.
On-Time Performance (OTP)	Percentage of completed flights meeting the applicable 15-minute schedule threshold.
Late Aircraft	Delay associated with the incoming aircraft arriving late from its previous leg, creating a potential ripple effect.
Air Carrier Delay	Delay attributed to airline operational factors such as crew, maintenance, baggage, cleaning, or related operations.
National Air System (NAS)	Delay associated with air-traffic-control, airspace, congestion, and broader system constraints.
Weather Delay	Delay directly attributed to weather conditions.
Security Delay	Delay attributed to security-related incidents or processes.
Cancellation	A scheduled flight that did not operate and is assigned one recorded cancellation reason.
Diversion	A flight that operates but lands somewhere other than its scheduled destination.

4. Dataset & Data Model

The project uses three CSV tables: **flights**, **airlines**, and **airports**. The documented model treats flights as the central fact table and airlines and airports as dimensions.

Table	Role	Key relationship
flights	Central fact table; one row per scheduled flight operation	AIRLINE, ORIGIN_AIRPORT, DESTINATION_AIRPORT
airlines	Airline dimension	IATA_CODE → flights.AIRLINE
airports	Airport dimension / role-playing reference	IATA_CODE → origin and destination airport fields

The airport table is logically role-playing because it is referenced for both origin and destination. The data-model documentation recommends separate origin and destination airport dimension instances when two active relationships are needed in a BI model.

4.1 Data-Quality Interpretation

Delay-cause fields are conditionally populated for flights reaching the applicable delay threshold. Their blanks therefore represent 'not applicable' in the source structure rather than automatically indicating dirty or missing data. This distinction was preserved during interpretation of delay-cause totals.

5. Data Preparation & Power BI Modeling

- Imported the three source CSV tables into Power BI.
- Established relationships between the flights fact table and airline/airport dimensions.
- Prepared date and time fields for analysis.
- Prepared readable cancellation-reason descriptions.
- Created delay-duration categories for severity analysis.
- Created DAX measures for the dashboard KPIs and analytical visuals.
- Validated major KPI counts against exact dashboard results rather than relying on abbreviated card displays.
- Tested slicer behavior and visual readability after dashboard construction.

The final dashboard is based on the project's implemented measures. Where the original documentation describes a broader set of potential KPIs, this report only claims completion for the measures and visuals that are actually present in the delivered Executive Overview.

6. KPI Framework — Final Dashboard

KPI	Final dashboard meaning	Validated result
Total Flights	Total scheduled flight records in the selected context.	5,819,079
Delayed Flights	Flights classified as delayed under the project's implemented 15-minute delay convention.	2,086,896
On-Time Flights	Flights classified as on time in the implemented dashboard status logic.	3,627,112
Delay Rate %	Share of flights classified as delayed under the implemented dashboard logic.	35.9%
Cancelled Flights	Flights with CANCELLED = 1.	89,884
Avg Duration (min)	Average actual elapsed gate-to-gate flight duration for the dashboard context.	137.01

The project KPI specification contains additional metrics such as Arrival OTP, Departure OTP, average delay, cancellation rate, diversion rate, cause shares, taxi times, and ranking measures. Those metrics are part of the analytical framework but are not represented as completed executive KPI cards in the final dashboard.

7. Final Dashboard — Implemented Design

The final dashboard deliberately uses a compact set of visuals so that the page remains readable and business-focused. It does not attempt to implement every metric in the original requirements document.

7.1 KPI Cards

- Total Flights
- Delayed Flights
- On-Time Flights
- Delay Rate %

- Cancelled Flights
- Avg Duration (min)

7.2 Analytical Visuals

Visual	Question answered
Flight Volume & Delays by Month	How do flight volume and delayed-flight volume change across the year?
Top 5 Airlines by Delay Rate	Which airlines have the highest delay rates in the selected context?
Flight Delay Distribution by Duration	How severe are the delays by duration category?
Monthly Flight Delay Rate	How does delay rate change across months?
Top 5 Origin Airports by Delayed Flights	Where is delayed-flight volume concentrated?
Total Delay Minutes by Cause	Which recorded delay causes contribute most to total delay minutes?
Flight Cancellations by Reason	What recorded reasons account for cancellations?

7.3 Interactivity

- Month slicer for temporal filtering.
- Airline slicer for carrier-level filtering.
- Visual cross-filtering based on standard Power BI interactions.
- Top-5 visuals configured for readability without persistent scrollbars.
- Standardized visual titles and number formatting.
- Cleaned donut visuals to improve label readability.

8. Business Insights

8.1 Overall Operational Performance

The dashboard covers 5,819,079 flights. It records 2,086,896 delayed flights, 3,627,112 on-time flights, and 89,884 cancellations. The implemented overall delay rate is 35.9%.

This establishes delay management as the primary operational issue highlighted by the final dashboard. The project does not attempt to translate this operational volume directly into financial impact because financial and passenger-level data are outside scope.

8.2 Monthly Performance

Monthly flight volume remains relatively stable, generally around 0.43M–0.52M flights per month, while delay rate varies more noticeably. February and June are among the higher-delay-rate months at roughly 41%, while September and October are among the lower periods at roughly 29%.

These are observed temporal patterns. The dashboard does not establish that seasonality, weather, aircraft rotations, airline operations, or air-system constraints caused the monthly differences.

8.3 Airline Performance

The Top 5 Airlines by Delay Rate visual identifies NK, F9, HA, VX, and US, with the leading carriers in the selected group reaching roughly 40–48%.

The correct business interpretation is that airline-level delay performance differs materially in the selected context and warrants further investigation. The dashboard does not establish the underlying operational causes of those differences.

8.4 Airport Concentration

The Top 5 Origin Airports by Delayed Flights are ATL, ORD, DFW, DEN, and LAX. ATL and ORD each exceed roughly 100,000 delayed flights.

These are high-volume delayed-flight concentrations, not a ranking of the 'worst airports.' Airport reliability should be benchmarked using a normalized measure such as delayed flights divided by total flights.

8.5 Delay Severity

Delay category	Share of delayed flights
1–15 minutes	50.96%
16–30 minutes	18.73%
31–60 minutes	15.03%
61–120 minutes	9.81%
120+ minutes	~5.47%

The largest single category is 1–15 minutes at 50.96%, while approximately 49.04% of delayed flights fall above 15 minutes. Longer delays have greater potential to disrupt connections, aircraft rotations, crew schedules, and passenger journeys.

8.6 Delay Cause Analysis

Total recorded delay minutes rank as: **Late Aircraft**, **Air Carrier**, **National Air System**, **Weather**, and **Security**.

This indicates that the accumulated delay-minute burden is concentrated primarily in operational and system-related categories. The report uses 'recorded contributor' rather than 'cause' when describing these results because the dashboard is descriptive and does not establish causal relationships.

8.7 Cancellation Analysis

The dashboard records 89,884 cancellations. Weather represents 54.35% of cancellations, followed by Airline / Carrier at 28.11% and National Air System at 17.52%, with Security the smallest share.

The key strategic insight is that the dominant cancellation reason differs from the dominant accumulated delay-minute contributors. Therefore, delay reduction and cancellation resilience should be treated as related but distinct operational problems.

9. Business Recommendations

Priority	Evidence from dashboard	Recommendation
1. Late-aircraft operations	Late Aircraft is the largest recorded contributor to delay minutes.	Investigate turnaround time, aircraft rotations, schedule buffers, and propagation of delays from earlier flight legs.
2. Airline performance review	Top 5 airlines show materially different delay rates.	Review delay rate, severity, cancellations, delay causes, airport mix, and route mix together before drawing performance conclusions.
3. Airport prioritization	ATL, ORD, DFW, DEN and LAX lead delayed-flight volume.	Use these airports as investigation priorities, then benchmark them with normalized delay rates.
4. Delay-severity management	50.96% of delayed flights are 1–15 minutes; ~49% exceed 15 minutes.	Separate operational targets for minor, moderate, and severe delays; monitor 61–120 and 120+ minute categories closely.

Priority	Evidence from dashboard	Recommendation
5. Weather contingency planning	Weather accounts for 54.35% of cancellations.	Strengthen disruption forecasting, passenger communication, rebooking, standby capacity, and aircraft/crew contingency planning.
6. Air-system monitoring	NAS is one of the largest recorded contributors to delay minutes.	Investigate recurring congestion by airport, route, and time period and incorporate it into capacity planning.
7. Recurring monitoring	Operational performance varies by time, airline, airport, and cause.	Convert the analysis into a recurring weekly/monthly monitoring process with refreshable KPIs and normalized benchmarks.

10. Limitations & Analytical Boundaries

- The dataset represents calendar year 2015 and does not describe current airline networks or operating conditions.
- The analysis is descriptive/diagnostic; it identifies patterns and concentrations rather than proving causation.
- Raw delayed-flight counts are influenced by flight volume. Airport and airline benchmarking should use normalized rates.
- Passenger-level, revenue, cost, and compensation data are unavailable, so financial and customer-impact values cannot be quantified.
- The project does not include predictive modeling or machine-learning forecasting.
- Delay-cause fields are conditionally populated in the source data and should be interpreted as not applicable when blank under the documented rules.
- The final dashboard is an Executive Overview implementation. The broader four-page design in the original Dashboard Requirements is treated as a future-extension specification, not as a claim about the delivered PBIX.
- The exact delay-rate interpretation must remain tied to the implemented dashboard measure and its documented 15-minute convention; future dashboard pages should not introduce a different denominator or delay definition.

11. Project Reconciliation & Final Implementation Statement

A final reconciliation was performed between the project documentation and the dashboard that was actually built. The final report intentionally uses the implemented dashboard as the delivery boundary.

Area	Final project position
Data source	Maven Analytics Airline Flight Delays dataset, U.S. DOT 2015 reporting.
Data model	Flights fact table with airline and airport reference dimensions; airport acts as a role-playing dimension.
Dashboard scope	One focused Executive Overview page in the final delivered dashboard.
KPI set	Six executive KPI cards actually implemented.
Analytical visuals	Seven visuals actually implemented and validated.
Filters	Month and Airline slicers actually implemented and tested.
Business analysis	Monthly, airline, airport, delay-severity, delay-cause, and cancellation analysis based on final visuals.
Recommendations	Derived from observed dashboard patterns and explicitly qualified where causation cannot be established.

Area	Final project position
Future extension	Detailed airline, airport/route, and delay-cause drill-down pages can be added later if required.

This implementation statement is important for interview use: it avoids claiming that the final PBIX contains pages or KPIs that were specified in the original requirements but were not part of the delivered Executive Overview.

12. Conclusion

The Airline Flight Operations & Delay Analysis project demonstrates an end-to-end data analytics workflow: understanding a business problem, documenting the domain, modeling a relational dataset, preparing data in Power BI, creating DAX-based KPIs, validating results, designing an executive dashboard, extracting business insights, and translating those insights into operational recommendations.

The final dashboard shows that the 2015 flight dataset contains a substantial delay-management challenge, with operational and system-related categories accounting for most recorded delay minutes and weather dominating cancellations. Airline and airport comparisons provide useful prioritization signals, but normalized rates and deeper operational context are required before making performance judgments.

The delivered project is therefore best positioned as a **descriptive and diagnostic BI case study**. Its next logical extension would be deeper normalized benchmarking, route-level analysis, time-of-day analysis, and predictive or financial-impact analysis if additional data becomes available.

13. GitHub Deliverables

Recommended artifact	Purpose
README.md	Recruiter-facing project overview, business problem, tools, dashboard screenshot, key findings and links.
01_Business_Requirement_Document.pdf	Business context, objectives, scope, stakeholders and business questions.
02_Domain_Document.pdf	Airline process and terminology.
03_Data_Dictionary.pdf	Table, column, key and data-quality definitions.
04_ER_Diagram.pdf	Data-model relationships and schema.
05_KPI_Document.pdf	Metric definitions and formulas.
06_Dashboard_Requirements.pdf	Original dashboard design specification / future-extension blueprint.
07_Insights_and_Recommendations.pdf	Detailed Phase 4 and Phase 5 analysis.
Airline_Flight_Operations_FINAL_Report.pdf	Integrated final project case study.
Airline_Flight_Operations.pbix	Final Power BI dashboard file.
screenshots/dashboard.png	Static portfolio preview of the final dashboard.

Repository note: For maximum consistency, the original Dashboard Requirements document should be labeled as the original design specification or updated to clearly distinguish planned scope from final implemented scope. This final report itself reflects the actual delivered dashboard.